# VaxImpact Simulator - Teacher Quick Guide

Website overview, classroom goals, student workflow, and facilitation notes

## 1. GOALS OF THE RESOURCE

VaxShield's mission is to make public health and scientific reasoning more accessible by connecting real-world health challenges with the science students learn in the classroom. VaxImpact is designed to give students an opportunity to investigate a phenomenon, work with a computational model, collect and analyze evidence, and use that evidence to explain a relationship.

The activities emphasize NGSS 3-dimensional learning through Science and Engineering Practices such as Developing and Using Models, Planning and Carrying Out Investigations, Analyzing and Interpreting Data, and Engaging in Argument from Evidence. The lesson also connects to LS2.D: Social Interactions and Group Behavior and the Crosscutting Concept of Cause and Effect.

***Teachers can request an Answer Key for FREE by emailing vaxshieldteam@gmail.com.***

## 2. KEY FEATURES OF THE VAXIMPACT WEBSITE

| Feature / location | What is included + how students use it |
|---|---|
| Main simulation panel - Center of the website titled "Disease Transmission Simulation" | The visual representation of the disease-transmission model. Students can watch the modeled outbreak unfold and observe how changing variables affects the spread. |
| Simulation controls – Right of the Main Simulation Panel | Six variables can be adjusted: vaccination rate, transmission rate, vaccine effectiveness, recovery time, mortality rate, and population density. Part I encourages exploration; Part II changes only vaccination rate. |
| Start - within the simulation controls | Starts the model. Students should allow a trial to finish before recording the final results in the worksheet. |
| Speed controls - At the top below Start | These controls appear once the simulation has started and allow students to speed up or slow down the run. They change the simulation speed, not the experimental settings. |
| Epidemic Timeline - Below the Simulation | After the model starts, students can scroll down to see the outbreak results over time. The worksheet asks students to record peak infections, total infections, and infection rates among vaccinated and unvaccinated people which can all be found here. |
| Learn More - Upper-Right Corner above the Main Simulation Pane | Provides additional information about the variables. Students can use it when a control is unfamiliar or when they want to understand how it influences the modeled outbreak. |

## 3. HOW THE WEBSITE CONNECTS TO THE WORKSHEET

| Worksheet stage | What students are doing |
|---|---|
| Part I - Explore & Question | Explore one variable at a time, record observations, write a testable question, and identify the independent variable(s), dependent variable(s), and controls. |
| Part II - Controlled Investigation | Use fixed baseline settings and compare 65%, 80%, and 95% vaccination. Students run three trials at each level and record the same four outcome measures. |
| Part III - Analyze | Use the completed averages to graph vaccination rate against average total infected and describe the trend shown by the data. |
| Part IV - Predict | Estimate what might happen at 90% vaccination and compare the estimates with the 85% and 95% benchmark values. |
| Part V - Analysis | Apply evidence to questions involving vaccine effectiveness, indirect transmission, individual vs. community effects, hospital burden, isolation, group mixing, and multiple variables. |

# Facilitation & Activity Options

## 4. TWO CLASSROOM ACTIVITY OPTIONS

### Simulator-Based Worksheet

The simplest option to administer. Students can work individually, in pairs, in small groups, or at home. Educators primarily facilitate. Students need a computer/device, internet access, the simulator, and the worksheet. No additional classroom materials are required. Printable Version and Editable Online Version are both available on the website.

### Interactive Outbreak Case Study

The more collaborative option. Students begin with a brief hands-on contact-transmission demonstration, take the role of a Student Public Health Task Force investigating a fictional school outbreak, analyze VaxImpact evidence, and prepare a one-minute response to a simulated school board. This version requires more preparation and facilitation but adds teamwork, scientific communication, and public-health decision-making.

### Which should I use?

Use the simulator-based worksheet for a simpler simulator and data-analysis experience or when class time is limited. Use the case study when you want a longer, collaborative, scenario-based lesson that extends the simulation into communication and decision-making.

## 5. BEFORE CLASS

Open the simulator before the lesson and run a quick test so you are familiar with the controls, Play, speed controls, and Epidemic Timeline. Students do not need prior experience with epidemiological modeling; the worksheet provides the experimental parameters and directions. For an online or homework assignment, make sure students have access to the simulator, worksheet, and graphing link. For an in-person lesson, consider demonstrating one short simulation before students begin Part I.

## 6. WHAT TO EMPHASIZE AS STUDENTS WORK

| Student action | Suggested teacher emphasis |
|---|---|
| Testable question | Independent variable = what changes. Dependent variable = what is measured. Controls = what remains constant. |
| Prediction | Have students explain the reasoning behind the increase/decrease prediction before they see the experimental results. |
| Repeated trials | The model can produce variable outcomes; students should look for a pattern across trials rather than rely on one unusual run. |
| Graphing | Recommended relationship: vaccination rate on the x-axis and average total infected on the y-axis. Students should label axes, units, and title. |
| Interpreting model output | A pattern in the model is evidence about the modeled system, not a guarantee of the exact same numerical outcome in a real community. |
| Part V | Use the analysis questions for discussion, writing, small-group work, or extension across more than one class period. |

## 7. MODEL LIMITATIONS / SAFETY

### Important

VaxImpact is a simplified teaching model. Its numerical outputs should not be used to determine a real-world herd-immunity cutoff or as medical guidance. The goal is to help students investigate relationships among variables, analyze evidence, and think critically about what a model can and cannot represent.

Website: http://vaxshield.org

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